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$TESS_{v2}$ magnitudes from Gaia DR3 XP spectra.

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Context: In the era of large-scale surveys delivering millimagnitude relative precision and discovering thousands of exoplanets and binary systems, a reassessment and improvement of the absolute flux scale is mandatory. Absolute flux calibration of TESS photometry underpins the characterisation of exoplanets and stellar variability, but TESS Input Catalog (TIC) magnitudes are uncertain by ≈ 0.15 mag. *Methods:* We derive synthetic $TESS_{v2}$ magnitudes by integrating Gaia DR3 XP spectra (produced by merging the Blue Photometer (BP) and Red Photometer (RP) spectra) over the official $TESS_{v2}$ passband, with the absolute scale set by the Gaia, SPSS, and CALSPEC libraries. Because the passband extends beyond the XP range (330–1050 nm), we compute magnitudes in a passband truncated at 1020 nm ($TESS_{v2,cut}$) and correct them to the full band: empirically for library stars and, for highly reddened and intrinsically red stars (e.g. M dwarfs), with ridge polynomial regression on BT-Settl and NextGen model grids in colour and extinction. *Results:* The zero points are 22.1911 mag ($TESS_{v2,cut}$) and 22.2027 mag ($TESS_{v2}$). For CALSPEC stars with $6 < TESS_{v2} < 16$ mag and $BP - RP < 5$ mag, XP-based and high-resolution $TESS_{v2,cut}$ magnitudes differ by a median of 0.0027 mag, with a standard deviation $\sigma = 0.0050$ mag (SPSS: 0.0039 and 0.0134 mag); the conversion to the full band adds 0.002 mag. The total uncertainty is about 5 mmag for blue sources and about 8 mmag for red or reddened sources. For 5899 TOIs, the new magnitudes differ from the TIC by -0.024 mag on average ($\sigma = 0.057$ mag). *Conclusions:* To our knowledge this is the first determination of TESS magnitudes from XP spectra; the precision of a few mmag for blue stars is well below the TIC uncertainty.

KEYWORDS:
Methods: data analysis — stars: general — Techniques: photometric — Techniques: spectroscopic — catalogs — surveys

1 | INTRODUCTION

Accurate absolute flux calibration — the conversion of astronomical signals into physical flux units tied to a standard reference system — is essential to fully exploit the scientific potential of exoplanet surveys. It enables several critical capabilities. First, absolute calibration permits the direct measurement of a planet's emitted or reflected flux relative to its host star, enabling precise measurements of planetary geometric albedos. Albedo provides valuable information on atmospheric properties, cloud coverage, and surface characteristics, helping to distinguish between different planet types. Second, uniform absolute calibration across the survey ensures consistent data quality for planets orbiting a wide range of stellar types and environments. This homogeneity is indispensable for statistical studies of atmospheric trends, metallicity gradients, and the influence of stellar irradiation or migration history. Third, accurate stellar flux calibration improves the detection and

characterization of stellar variability (spots, flares, and pulsations) and helps identify residual instrumental systematics such as jitter or persistence effects.

The NASA Transiting Exoplanet Survey Satellite (TESS; Ricker et al., 2016), launched in April 2018, surveys the sky in a broad photometric bandpass covering approximately 534–1060 nm (at 10% peak transmission), with maximum response near 800 nm and a red tail extending to about 1100 nm. The TESS Input Catalog (TIC) provides synthetic TESS magnitudes (T_{mag}) that are derived from Gaia DR2 photometry using a colour-dependent polynomial transformation (Paegert et al., 2021; Stassun et al., 2019). This relation was constructed using PHOENIX model atmospheres (Husser et al., 2016) for dwarfs, subgiants, and giants across a wide range of metallicities. The transformation is given for stars with $-0.2 < \mathrm{BP} - \mathrm{RP} < 3.5$, but its accuracy degrades for stars with $\mathrm{BP}-\mathrm{RP} \gtrsim 2$ mag. Stassun et al. (2019) estimated that the typical uncertainty in the derived TIC T_{mag} values is at most ∼0.15 mag for the majority of stars. A more robust absolute calibration framework will facilitate the combination of TESS data with existing and future observations, maximising the overall scientific return of the mission.

In this work, we derive improved synthetic TESS ($\mathrm{TESS_{v2}}$) magnitudes from Gaia DR3 XP spectra using the TESS passband version 2 (v2). XP refers to the low-resolution prism spectra obtained with the Blue Photometer (BP) and Red Photometer (RP) of Gaia. The absolute calibration is established using the Gaia spectrophotometric standard stars (SPSS) and the CALSPEC libraries (Bohlin, Deustua, & de Rosa, 2019; Pancino et al., 2021), which define the absolute flux scale adopted for the Gaia mission. For intrinsically red sources and highly reddened sources, relative magnitude corrections as a function of observed colors are estimated with stellar models.

The structure of the paper is as follows. The TESS passband version 2 (v2) and its zero points are described in Sect. 2, respectively. Synthetic $\mathrm{TESS_{v2}}$ magnitudes computed from high-resolution SPSS and CALSPEC spectra are presented in Sect. 3. In Sect. 4, we introduce a method to derive $\mathrm{TESS_{v2}}$ magnitudes from Gaia XP spectra and evaluate its accuracy and precision. In Sect. 5 the impact of the new calibration on science is discussed. Eventually, in Sect. 6, a summary is given.

2 | THE TESS PASSBAND AND ZEROPOINTS

The Gaia DR3 release made available, for the first time, low-resolution XP spectra for more than 200 million sources, enabling synthetic photometry in virtually any user-defined optical passband (Gaia Collaboration et al., 2023). The spectrophotometric calibration of these spectra is based on a network of flux standards spanning a wide range of spectral types and magnitudes, and has been demonstrated to achieve internal consistency at the few-millimagnitude level across most of the optical range. Convolving the calibrated Gaia XP spectra with a target transmission curve it is possible to predict magnitudes in any passband covered by the Gaia spectral range, providing a powerful bridge between the Gaia photometric system and other ground- or space-based surveys. In this context, the TESS passband represents a natural application of this approach, given the large number of stellar targets observed by both missions.

The official NASA v2.0 TESS passband[1], released by Dr. Roland Vanderspek (2020) at the MIT Payload Operations Center, is available in tabular form from the NASA TESS Science Support Center, see also the online NASA documentation and the description of the earlier version presented by Ricker et al. (2015). The TESS passband is not included among the bandpasses supported by the official Gaia DR3 synthetic photometry, nor by any earlier Gaia release. The GAIAXPY tool (Gaia Collaboration et al., 2023) allows user-defined synthetic photometry, but only for bandpasses falling within the Gaia XP spectral coverage (330–1050 nm). The TESS response curve spans approximately 534–1060 nm, and its red tail extends beyond the Gaia XP range, as shown in Fig. 1. With a TESS filter cut at 1020 nm (the default Gaia maximum sampling wavelength), we generated a new filter ($\mathrm{TESS_{v2,cut}}$), which covers the entire Gaia XP spectrum. The fraction of stellar flux beyond 1020 nm is only a few percent, so we investigated whether an empirical absolute calibration of TESS magnitudes could still be achieved using a truncated passband and the Gaia XP spectra.

In order to calculate the $\mathrm{TESS_{v2}}$ zeropoints, we adopted the photon-counting detector formalism following Bessell and Murphy (2012), and the same rescaled flux-calibrated CALSPEC spectrum of α Lyr as used in Gaia DR3[2]. Precisely, Riello et al. (2021) adopt the Vega spectrum `alpha_lyr_mod_002.fits` from the CALSPEC Calibration Library (2013–2015 release), rescaled to a flux density of 3.62286×10^{-11} W m^{-2} nm^{-1} at $\lambda = 550.0$ nm, which corresponds to the adopted flux of an unreddened A0V star with $V = 0$ mag. We summarize the calculated $\mathrm{TESS_{v2}}$ zeropoints in Table 1.

[1]This response function is based on the CCD quantum efficiency measurements and measurements of the optical properties of the lens glasses (private communication by Dr. Schanche Nicole E.).

[2]Gaia DR3 documentation.

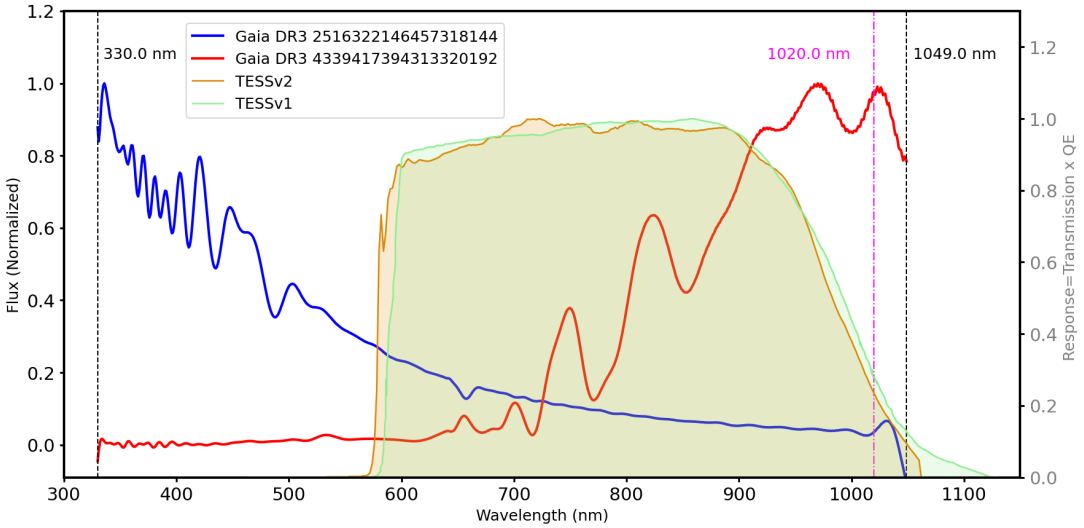


FIGURE 1 XP sampled spectra of Gaia DR3 1651137131123978112 (blue) and that of Gaia DR3 6324325225803432320 (red), which are spectrophotometric standards from the CALSPEC library. Vertical black-dashed lines mark the two wavelength endpoints of the spectrum at 330 nm and 1050 nm. The vertical magenta-dashed line marks the maximum wavelength of the Gaia default sampling (1020 nm). Overplotted are the TESS v1 (light green) passband by Barclay (2018) (Ricker et al., 2015) and the TESS v2 (golden orange) passband by Vanderspek (2020). The red tail of the TESS filter exceeds the XP spectral range.

TABLE 1 Zeropoints for synthetic photometry in the VEGAMAG system (flux densities in erg s^{-1} cm^{-2} Å^{-1}; add 5 mag to convert to W m^{-2} nm^{-1}). The rescaled file `alpha_lyr_mod_002.fits` is used as for the Gaia filters (Riello et al., 2021).

Origin	$ZP_{TESS_{v2,cut}}$	$ZP_{TESS_{v2}}$
VEGA	22.1911	22.2027

3 | $TESS_{V2}$ MAGNITUDES: VALIDATION SAMPLE FROM HIGH-RESOLUTION SPECTRA

3.1 | CALSPEC

Synthetic $TESS_{v2}$ and $TESS_{v2,cut}$ magnitudes were calculated using the high-resolution spectra from the CALSPEC and SPSS libraries and the zeropoints derived in the previous section. The synthetic magnitudes were computed with the PyPhot package[3].

The CALSPEC library contains the ≈ 140 STIS spectra that are flux standards on the HST system (Bohlin et al., 2019), (CALSPEC link). Following Munaretto et al. (2025), for flux density validation only a subsample of library stars was used, specifically those stars that had Gaia DR3 parameters phot_variable_flag ! = "VARIABLE", and non_single_star=0, RUWE< 1.5, and contamination in their flux measures C^* < 0.1. Only 87 stars satisfy these criteria and are retained, **and 58 are fainter than 6 mag (removing the saturated) and brighter than 16 mag (stars fainter than that are affected by hockey-stick effect).** However, the colour range of the CALSPEC sample is limited; the bulk of the CALSPEC stars has BP − RP below 1.5 mag, with only two stars at BP − RP ≈ 3 and ≈ 5 mag. **It is important to state that the phot_variable_flag ! = "VARIABLE" does not mean that the star does not vary, but that variability was not detected.**

We compute rp_mag_err as the uncertainty on the mean RP magnitude. By error propagation, $\sigma_mag = 1.0857/phot_rp_mean_flux_over_error$**. This is the standard error of the mean over all the epochs Gaia observed. The rp_mag_scatter_est estimates epoch-to-epoch scatter,** $\sigma_epoch = rp_mag_err \times sqrt(phot_rp_n_obs)$**. The mean of all rp_mag_scatter_est is 0.0048 mag with a standard deviation of 0.0039 mag. Three stars only have an rms of 0.021, 0.017, and 0.015 mag, i.e. 5-7 times than the median scatter (0.0034 mag). The rp_mag_scatter_est is a standard deviation in magnitudes and approximates how much a single RP measurement of that star typically deviates from its**

[3] A python code, PYPHOT, freely distributed by Dr. Fouesneau. The code has been tested and used already by several groups, for example, see the work of Lahén, Naab, and Kauffmann (2022).

mean. It is caused by photometric noise and real variability, it is therefore an upper limit on the variability in RP. Gaia samples each source a few tens of times over about 34 months, so it cannot resolve variability on short timescales.

We also retrieved the TESS SPOC 2-min light curves (PDCSAP flux, QUALITY = 0, 5σ clipping). For each sector we computed the 5th–95th percentile range of the magnitude distribution, Δmag(SPOC), and kept the sector with the largest value. Of the 58 CALSPEC best stars, 53 have both a SPOC 2-min light curve and an RP scatter estimate. The median Δmag(SPOC) is 0.012 mag, and 25 stars (47%) have Δmag(SPOC) $\leq$ 0.010 mag. Converting Δmag(SPOC) to a Gaussian-equivalent dispersion (Δmag(SPOC)/3.29), the median TESS value is 0.0036 mag, close to the median Gaia rp_mag_scatter_est of 0.0034 mag. The two quantities are correlated (Spearman $\rho = 0.60$), and for 42 of 53 stars they agree within a factor of 2. Four stars show Δmag(SPOC)$>$ 0.1 mag. Because Δmag(SPOC) includes photometric noise and is taken at the most variable sector, it is an upper limit on the intrinsic variability. For most of the sample the variability is therefore at or below the epoch-to-epoch scatter already included in our error budget. Since the stars are independent one to another, the mean zero-point is not biased.

In Fig. 2, the synthetic $TESS_{v2}$ magnitudes from the high-resolution CALSPEC spectra are compared with the TIC magnitudes[4]. The TIC T_{mag} values appear accurate only within 0.10-0.15 mag overall, consistently with the verification of Stassun et al. (2019).

3.2 | SPSS

The spectrophotometric standard stars (SPSS) is a library of spectra specifically designed for the Gaia calibration, it includes 169 stars (Pancino et al., 2021). By using the same Gaia quality flags as done for the CALSPEC sample, a subsample of 129 library stars is used. **The quality of the SPSS library is that extends the colour range from $\approx$ 1.5 mag to $\approx$ 3 mag. However, the synthetic magnitudes extracted from ground-based data remains more that twice noisier than that from Hubble.**

4 | $TESS_{V2}$ MAGNITUDES FROM XP SPECTRA

For the CALSPEC library stars, synthetic $TESS_{v2,cut}$ magnitudes were derived from the XP spectra. Gaia XP spectra enable us to derive $TESS_{v2,cut}$ magnitudes with excellent internal consistency. For $6 < TESSv2_cut < 16$ mag and mostly BP $-$ RP $<$ 1.5 mag, the median and standard deviation of the differences between the $TESS_{v2,cut}$ magnitudes from the high-resolution CALSPEC spectra and those from the XP spectra are 0.0027 mag and 0.0050 mag, as shown in Fig. 3. If we use the $TESS_{v2}$ magnitudes from the high-resolution spectra (original filter), the mean difference is +0.0018 mag and the standard deviation remains 0.0058 mag. There is no evident trend with color, but the BP$-$RP colour range of the CALSPEC sample is limited (mostly BP $-$ RP $<$ 1.5 mag).

The synthetic $TESS_{v2,cut}$ magnitudes derived from the high-resolution SPSSV3.4 spectra differ from the $TESS_{v2,cut}$ magnitudes computed from the Gaia DR3 XP spectra by a median offset of 0.0039 mag, with a standard deviation of $\sigma = 0.0134$ mag. For the SPSSV3.4, the scatter is more than twice that of the CALSPEC spectra. **Indeed, Munaretto et al. (2025) relied exclusively on the CALSPEC library for the absolute calibration of the BepiColombo/SIMBIO-SYS camera. We retain the SPSS sample here because it independently yields the same first-order slope for the $TESS_{v2,cut}$ to $TESS_{v2}$ correction as the CALSPEC stars (Sect. 4.1.2). Therefore, it supports our calibration.**

4.1 | Red-tail correction

Our adopted procedure for deriving $TESS_{v2}$ magnitudes is as follows:

- i) generate the $TESS_{v2,cut}$ magnitudes from XP spectra;
- ii) apply a correction for the missing spectral coverage.

The bulk of the stars in the CALSPEC and SPSSV3.4 libraries are at BP $-$ RP $<$ 1.5 mag, which corresponds to spectral types earlier than M4–M5 stars (see Table 2); the color range does not cover late M stars (which are important in the exoplanet field of study). Furthermore, interstellar extinction changes the shape of the stellar energy distribution, further complicating the retrieval of the true correction. The library stars are at low extinction; for example, the CALSPEC library includes stars with reddening in B-V, E(B-V), $<$ 0.12 mag ($A_V \approx$ 0.37 mag and $A_{K_s} \approx$ 0.03 mag) (Munaretto et al., 2025). We, therefore, estimate the correction with simulations. We build a model of corrections from $TESS_{v2,cut}$ to $TESS_{v2}$ using theoretical model atmospheres and an assessment of interstellar extinction effects.

4.1.1 | Theory: a correction model

We recalculated the correction using synthetic spectra in order to cover a wider colour range.

[4]The TIC magnitudes are taken from the MAST catalog, using astroquery.mast and the Gaia DR2 source_id.

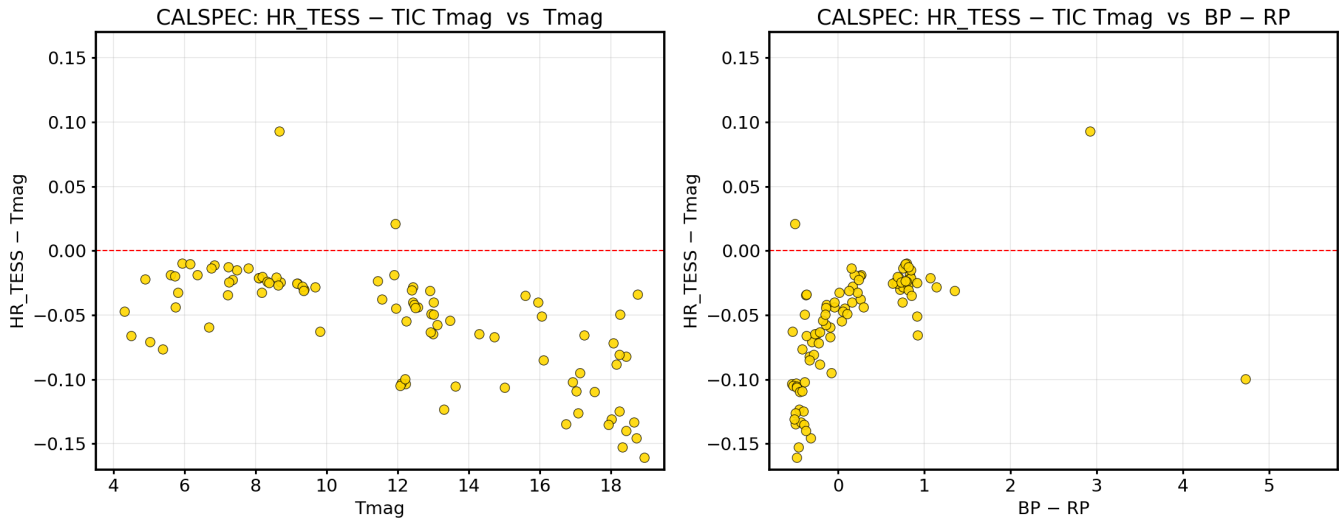


FIGURE 2 CALSPEC stars. *Left Panels:* The differences between the synthetic $TESS_{v2}$ magnitudes inferred from the CALSPEC spectra and the TIC magnitudes are plotted versus the TIC magnitudes. *Upper-Right Panels:* The same differences (as in the left panel) are plotted versus the BP − RP colours.

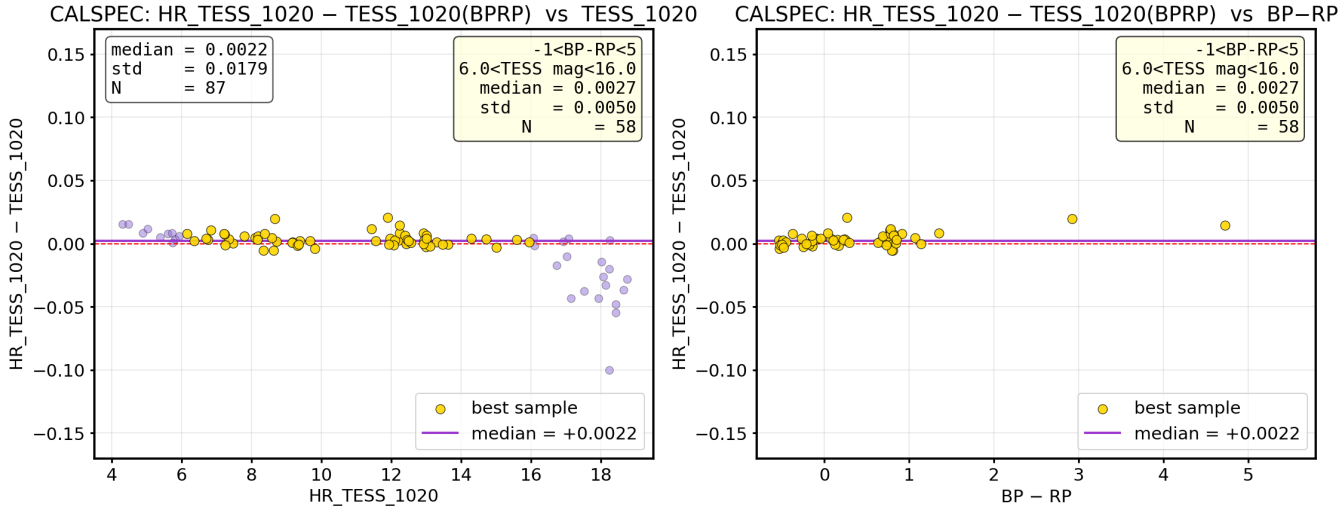


FIGURE 3 The differences between the synthetic $TESS_{v2,cut}$ magnitudes derived from the high-resolution CALSPEC spectra (HR_TESS_1020) and the synthetic $TESS_{v2,cut}$ magnitudes from the XP spectra (TESS_1020) are plotted versus the $TESS_{v2,cut}$ magnitudes (*left panel*) and versus the Gaia XP colours in the (*right panel*).

We employed the NextGen and BT-Settl families of PHOENIX model atmospheres developed by Allard and collaborators (Allard, 2014; Allard et al., 2011)[5]. NextGen is an older gas-phase-only grid, while the more recent BT-Settl models incorporate dust-cloud physics and updated solar abundances. BT refers to the Barber & Tennyson BT2 water-vapour line list; Settl denotes the improved treatment of cloud formation, sedimentation and gravitational settling of dust grains. The NextGen models are known to overestimate $T_{\rm eff}$ as a function of $J - K_s$ along the lower main sequence (Allard et al., 2011). In contrast, BT-Settl colours agree reasonably well with most empirical $T_{\rm eff}$ scales across the stellar and substellar (M–L–T) regimes.

We used BT-Settl models at solar metallicity with $T_{\rm eff}$ = 34 000–800 K ($\log g = 4.0$) and $T_{\rm eff}$ = 5000–800 K ($\log g$ =

[5]The spectra were retrieved from the Virtual Observatory SED Analyzer (Bayo et al., 2008). Both the BT-NextGen_agss2009 (gas-only) and BT-Settl (gas+dust) grids were used.

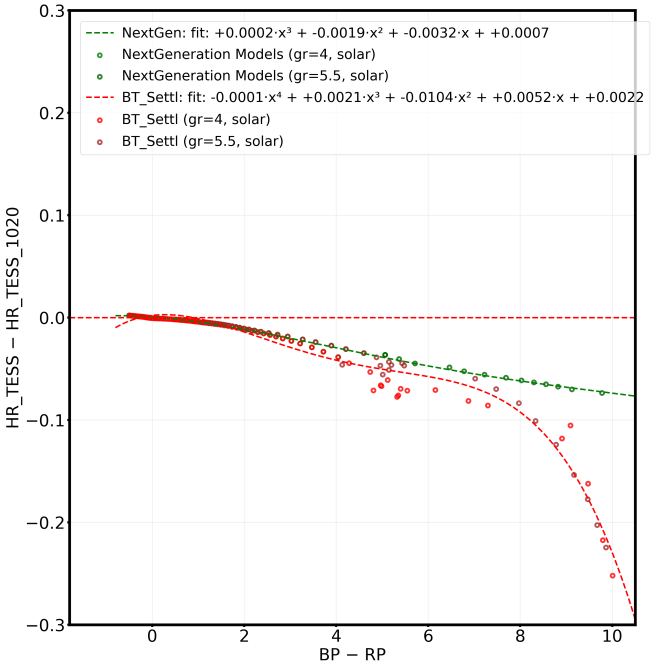


FIGURE 4 The differences of theoretical synthetic $TESS_{v2}$ magnitudes minus the synthetic $TESS_{v2,cut}$ magnitudes are plotted with open circles. The models from the NextGen grid at solar metallicity by Allard et al. (2011) are in green (green indicates gravity 4 and darkgreen gravity 5.5). The models from the BT-Settl grid at solar metallicity by Allard et al. (2011) are in red (red indicates gravity 4 and darkred gravity 5.5). The dotted lines are fit to the datapoints at gravity 4 and 5.

5.5). For the BT-NextGen_agss2009 grid we adopted $T_{\rm eff}$ = 34 000–2600 K ($\log g$ = 1.5 and 4.0) and $T_{\rm eff}$ = 5000–800 K ($\log g$ = 5.5). Spectra were rebinned to ≈ 9000 wavelength points before synthetic photometry was computed with PyPhot. The resulting $TESS_{v2} - TESS_{v2,cut}$ differences are shown versus BP − RP in Fig. 4. Table 2 lists the colours and corrections for cool dwarfs ($\log g = 5.5$, $A_K = 0$ mag). For M dwarfs (2600–3900 K) the two model families yield essentially identical corrections; they diverge for L and T dwarfs.

Interstellar extinction also affects the amount of correction. To quantify its impact, we reddened the models. We reddened the model spectra with the optical extinction law by Cardelli, Clayton, and Mathis (1989), extrapolated into the near-infrared with a power-law index of −2.1 (Messineo, 2025). We applied reddening in steps of A_{K_s} = 0.0, 0.2, 0.4, 0.6, 0.8, 1.0, 1.2, and 1.4 mag (corresponding to A_V = 0.0, 2.6, 5.2, 7.8, 10.4, 13.0, 15.6, and 18.2 mag).

At fixed BP − RP, the zero-extinction corrections $\Delta = TESS_{v2} - TESS_{v2,cut}$ decrease on average by 0.0140 mag ($\sigma = 0.0023$) for $A_{K_s} = 0.2$ mag, 0.0307 mag ($\sigma = 0.0040$) for $A_{K_s} = 0.4$ mag, and so on up to 0.1530 mag ($\sigma = 0.0118$) for $A_{K_s} = 1.4$ mag (NextGen, $\log g = 5.5$). The mean dependence can be approximated by

$$\Delta = -(0.03282\, A_{K_s}^2 + 0.06366\, A_{K_s}).$$

Interstellar extinction has a significant impact on the estimates Δ and must be modelled. Because higher-order polynomial terms in BP − RP and A_K are strongly multicollinear, unregularised least-squares fits become unstable and develop unphysical oscillations. We therefore adopted Ridge regression (L2 regularisation) on standardised polynomial features. Ridge regression yields a smooth, robust, and physically plausible correction surface while suppressing the unphysical oscillations that appear in pure ordinary least squares (OLS) solutions, while preserving the same cross-validated accuracy as OLS.

The model coefficients are listed in Tables 3 , 4 , 5 and 6 and enter the expansion

$$\Delta\mathrm{TESS} = c_0 + \sum_{i=1}^{X} \sum_{j=0}^{i} c_{i,j} \cdot \mathrm{BP\text{-}RP}^{i-j} \cdot A_K^j \qquad (1)$$

where the coefficients $c_{i,j}$ are expressed in the original (unscaled) variables and X is the polynomial degree. c_0 is the intercept. All fits were performed for BP − RP ≤ 10, except for the BT-Settl $\log g = 4.0$ grid, which is optimally described by a cut at BP − RP ≤ 5.

The polynomial degrees and the regularization strength α were selected by minimizing the cross-validation root-mean-square error (CV RMS). Features were standardized before the Ridge fit; α was then tuned on a logarithmic grid by repeated

shuffled 5-fold cross-validation (10 repeats). The final value of α was chosen according to the one-standard-error (1-SE) rule—the largest α whose CV RMS lies within one standard error of the minimum—rather than the absolute minimum itself. This choice yields a slightly more conservative model while remaining statistically indistinguishable from the CV optimum.

A fifth-degree polynomial is adopted for NextGen $\log g = 1.5$ ($\alpha = 1.0 \times 10^{-3}$) and for both BT-Settl grids ($\alpha = 1.0 \times 10^{-3}$ for $\log g = 4.0$ and $\alpha = 1.0 \times 10^{-4}$ for $\log g = 5.5$). For NextGen $\log g = 5.5$ a third-degree surface is sufficient ($\alpha = 3.2 \times 10^{-2}$). In the BT-Settl cases the CV RMS decreases only marginally beyond degree 4; the fifth degree was retained because it removes residual trends.

The resulting surfaces achieve CV RMS values of 0.0033 ± 0.0001 mag (NextGen $\log g = 1.5$), 0.0009 ± 0.0000 mag (NextGen $\log g = 5.5$), 0.0010 ± 0.0000 mag (BT-Settl $\log g = 4.0$) and 0.0058 ± 0.0002 mag (BT-Settl $\log g = 5.5$). These figures confirm that the systematic trends are recovered at the few-millimag level across the full colour and extinction range. Pure ordinary-least-squares fits yield essentially identical CV RMS values, but the modest regularisation provided by the 1-SE α suppresses coefficient oscillations and improves the stability of the correction outside the densest regions of the model grids.

The model results are shown in Figs. 5, 6, 7, and 8.

TABLE 2 Theoretical BP − RP colours of cool dwarfs (solar, $\log_{10}$(g)=5.5, and $A_K = 0.0$ mag), and corresponding $\mathrm{TESS_{v2}} - \mathrm{TESS_{v2,cut}}$ (Corr).

T_{eff} [K]	NextGen BP − RP [mag]	Settl BP − RP [mag]	NextGen Corr1 [mag]	Settl Corr2 [mag]	Obsfit Corr3 [mag]
4200	1.60	1.60	−0.0073	−0.0073	−0.0051
3900	1.82	1.82	−0.0090	−0.0090	−0.0062
3600	2.10	2.11	−0.0115	−0.0115	−0.0076
3200	2.72	2.72	−0.0166	−0.0166	−0.0108
2900	3.55	3.55	−0.0239	−0.0238	−0.0150
2600	4.60	4.60	−0.0347	−0.0347	−0.0204
2200	4.96	6.46	−0.0486	−0.0469	−0.0222
1900	5.43	7.69	−0.0587	−0.0444	−0.0246
1600	7.02	8.56	−0.0651	−0.0596	−0.0327
1200	8.78	10.59	−0.0781	−0.1240	−0.0417
900	9.67	13.42	−0.0986	−0.2026	−0.0462

– Corr1=NextGen $\mathrm{TESS_{v2}} - \mathrm{TESS_{v2,cut}}$

– Corr2=Settl $\mathrm{TESS_{v2}} - \mathrm{TESS_{v2,cut}}$

– Corr3= $\mathrm{TESS_{v2}} - \mathrm{TESS_{v2,cut}}$ This correction is calculated with the extrapolation to the fit to the synthetic magnitudes from observed XP data (CALSPEC and SPSSV3.4).

TABLE 3 Coefficients of the 5th-degree ΔTESS polynomial (NextGen $\log_{10}$(g)=1.5), fitted for BP−RP ≤ 10 mag. Features were standardized and $\alpha = 1.0e-03$ was chosen by repeated shuffled 5-fold cross-validation; coefficients are given in the original (unscaled) variables.

Term	Coefficient
Constant (c_0)	$2.371456e-04$
BPRP	$-4.469043e-03$
A_K	$3.329082e-03$
BPRP^2	$-8.995278e-04$
$\mathrm{BPRP} \times A_K$	$-1.913143e-02$
A_K^2	$3.569046e-02$
BPRP^3	$2.290123e-04$
$\mathrm{BPRP}^2 \times A_K$	$5.755135e-03$
$\mathrm{BPRP} \times A_K^2$	$-5.492851e-02$
A_K^3	1.720371e-01
BPRP^4	$-7.509053e-05$
$\mathrm{BPRP}^3 \times A_K$	8.068993e-04
$\mathrm{BPRP}^2 \times A_K^2$	$-2.746039e-03$
$\mathrm{BPRP} \times A_K^3$	$-2.995125e-02$
A_K^4	1.065885e-01
BPRP^5	$-1.075279e-05$
$\mathrm{BPRP}^4 \times A_K$	1.165779e-04
$\mathrm{BPRP}^3 \times A_K^2$	2.838964e-04
$\mathrm{BPRP}^2 \times A_K^3$	$-2.381658e-03$
$\mathrm{BPRP} \times A_K^4$	$-1.750769e-02$
A_K^5	9.891859e-02

4.1.2 | Empirical correction for libraries stars

For the library stars, which are blue sources, we also investigate the correction from $\mathrm{TESS_{v2,cut}}$ magnitudes to $\mathrm{TESS_{v2}}$ magnitudes with the high-resolution spectra. In Fig. 9, the differences are plotted versus the magnitudes and BP − RP colours. Both CALSPEC and SPSSV3.4 show the same linear correlation between the differences of the two sets of magnitudes and the BP − RP colours, as listed in Table 7. For red stars, the correction is not negligible; for example, for BP − RP = 5 mag, the $\Delta = -0.022$ mag.

Below BP − RP < 3 mag, the Δ from the observed data shown in Fig. 9 are indistinguishable from those theoretically derived (within 0.005 mag as shown in Table 2).

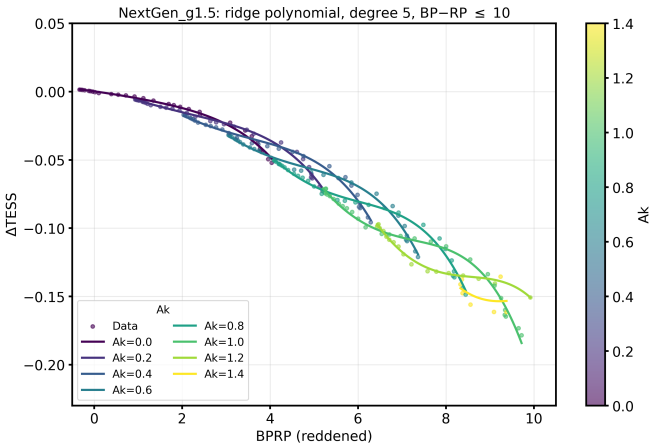


FIGURE 5 Differences between theoretical synthetic $\mathrm{TESS_{v2}}$ magnitudes and synthetic $\mathrm{TESS_{v2,cut}}$ magnitudes (filled circles). Models are taken from the Next Generation grid (Allard et al. (2011)) at solar metallicity, $\log_{10} g = 1.5$, and $T_{\rm eff}$ ranging from 2600 to 35000 K. The models have been reddened to $A_{\rm K_s}(A_V) = 0.0(0.0), 0.2(2.6), 0.4(5.2), 0.6(7.8), 0.8(10.4), 1.0(13.0), 1.2(15.6), 1.4(18.2)$ mag, respectively. Points corresponding to the same extinction are connected by fitted lines. Models and points are coloured according to the viridis colour map, scaled by A_{K_s}.

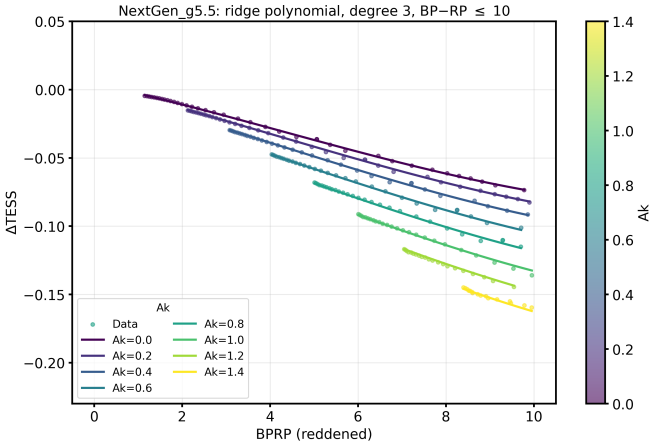


FIGURE 6 Differences between theoretical synthetic $\mathrm{TESS_{v2}}$ magnitudes and synthetic $\mathrm{TESS_{v2,cut}}$ magnitudes (filled circles). Models are taken from the Next Generation grid (Allard et al. (2011)) at solar metallicity, $\log_{10} g = 5.5$, and $T_{\rm eff}$ ranging from 800 to 5000 K. The models have been reddened to $A_K(A_V) = 0.0(0.0), 0.2(2.6), 0.4(5.2), 0.6(7.8), 0.8(10.4), 1.0(13.0), 1.2(15.6), 1.4(18.2)$ mag, respectively. Points corresponding to the same extinction are connected by fitted lines. Models and points are coloured according to the viridis colour map, scaled by A_{K_s}.

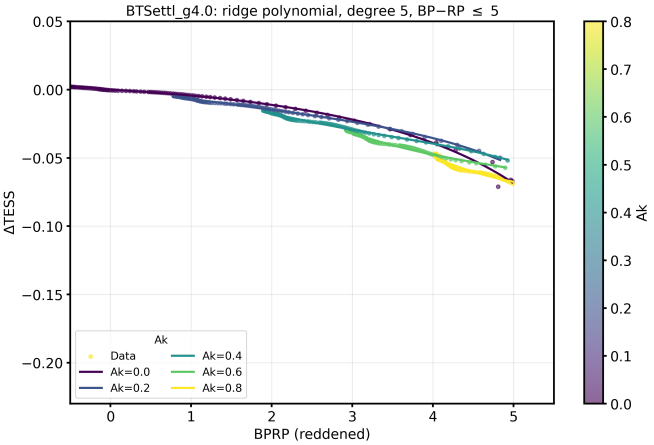


FIGURE 7 Differences between theoretical synthetic $\mathrm{TESS_{v2}}$ magnitudes and synthetic $\mathrm{TESS_{v2,cut}}$ magnitudes (filled circles). Models are taken from the BT Settl grid (Allard et al. (2011)) at solar metallicity, $\log_{10} g = 4.0$, and $T_{\rm eff}$ ranging from 800 to 34000 K. The models have been reddened to $A_K(A_V) = 0.0(0.0), 0.2(2.6), 0.4(5.2), 0.6(7.8), 0.8(10.4), 1.0(13.0), 1.2(15.6), 1.4(18.2)$ mag, respectively. Points corresponding to the same extinction are connected by fitted lines, however a good fit was obtained for points $\mathrm{BP} - \mathrm{RP} < 5$ mag. Models and points are coloured according to the viridis colour map, scaled by A_{K_s}.

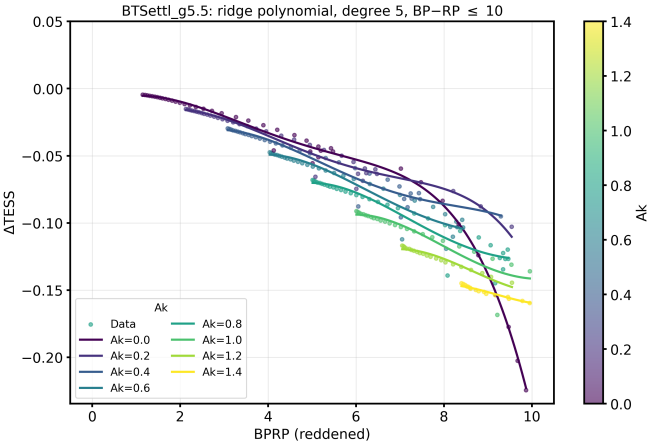


FIGURE 8 Differences between theoretical synthetic $\mathrm{TESS_{v2}}$ magnitudes and synthetic $\mathrm{TESS_{v2,cut}}$ magnitudes (filled circles). Models are taken from the BT Settl grid (Allard et al. (2011)) at solar metallicity, $\log_{10} g = 5.5$, and $T_{\rm eff}$ ranging from 800 to 5000 K. The models have been reddened to $A_K(A_V) = 0.0(0.0), 0.2(2.6), 0.4(5.2), 0.6(7.8), 0.8(10.4), 1.0(13.0), 1.2(15.6), 1.4(18.2)$ mag, respectively. Points corresponding to the same extinction are connected by fitted lines. Models and points are coloured according to the viridis colour map, scaled by A_{K_s}.

TABLE 4 Coefficients of the 3th-degree ΔTESS polynomial (NextGen $\log_{10}$(g)=5.5), fitted for BP−RP ≤ 10 mag. Features were standardized and $\alpha = 3.2e-02$ was chosen by repeated shuffled 5-fold cross-validation; coefficients are given in the original (unscaled) variables.

Term	Coefficient
Constant (c_0)	4.643040e-03
BPRP	-7.074736e-03
A_K	-5.303975e-04
BPRP^2	-4.064623e-04
BPRP × A_K	-3.616156e-03
A_K^2	-3.530954e-02
BPRP^3	3.214983e-05
BPRP^2 × A_K	3.610048e-05
BPRP × A_K^2	4.036175e-05
A_K^3	9.739784e-03

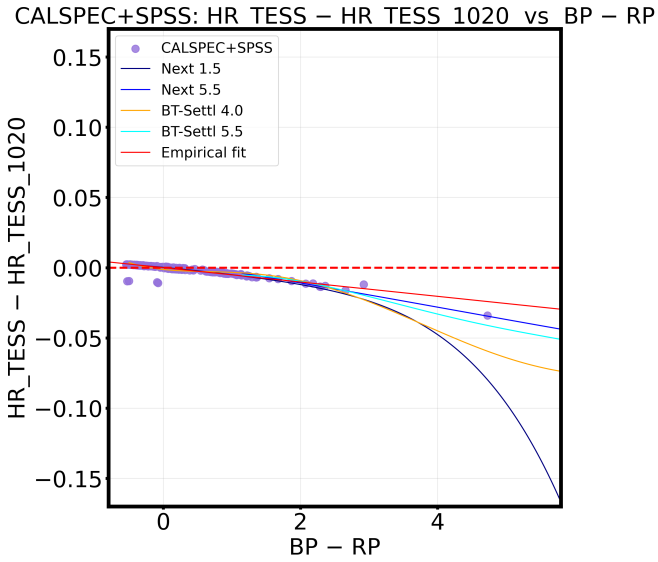


FIGURE 9 The differences between the synthetic $\mathrm{TESS_{v2}}$ magnitudes and the synthetic $\mathrm{TESS_{v2,cut}}$ magnitudes derived from the CALSPEC and SPSSV3.4 spectra are plotted versus the Gaia XP colours. The long-dashed red line mark the zero difference. The red line marks the empirical fit from the library stars, the blue, cyan, orange and navy curves show the theoretical predictions using the NextGen $\log_{10}$(g)=5.5, the BT_settl $\log_{10}$(g)=5.5, the BT_settl $\log_{10}$(g)=4.0, and the NextGen $\log_{10}$(g)=1.5 models, respectively. Each model is plotten within the validity range of the fit.

TABLE 5 Coefficients of the 5th-degree ΔTESS polynomial (BT-Settl $\log_{10}$(g)=4.0), fitted for BP − RP ≤ 5 mag. Features were standardized and $\alpha = 1e-03$ was chosen by repeated shuffled 5-fold cross-validation; coefficients are given in the original (unscaled) variables.

Term	Coefficient
Constant (c_0)	2.808332e-05
BPRP	$-3.674620e-03$
A_K	$-6.969206e-03$
BPRP^2	$-1.854357e-04$
BPRP × A_K	$-1.325329e-02$
A_K^2	2.590481e-02
BPRP^3	$-5.892869e-04$
BPRP^2 × A_K	5.838539e-03
BPRP × A_K^2	$-3.589737e-02$
A_K^3	1.091646e-01
BPRP^4	1.385153e-04
BPRP^3 × A_K	3.779117e-05
BPRP^2 × A_K^2	$-4.340996e-03$
BPRP × A_K^3	$-3.268350e-02$
A_K^4	1.592007e-01
BPRP^5	$-1.869873e-05$
BPRP^4 × A_K	1.676533e-05
BPRP^3 × A_K^2	9.746702e-04
BPRP^2 × A_K^3	6.702336e-03
BPRP × A_K^4	$-7.673233e-02$
A_K^5	1.657967e-01

4.2 | Error

The total uncertainty on a $\mathrm{TESS_{v2}}$ magnitude derived from XP spectra combines four terms, which we add in quadrature, assuming they are independent. (i) The residual scatter between the XP-based and the high-resolution $\mathrm{TESS_{v2,cut}}$ magnitudes for the CALSPEC sample ($6 < \mathrm{TESS_{v2}} < 16$, $-1 < \mathrm{BP} - \mathrm{RP} < 5$), $\sigma = 0.0050$ mag. (ii) The conversion from $\mathrm{TESS_{v2,cut}}$ to $\mathrm{TESS_{v2}}$, which adds about 0.002 mag for the library stars. (iii) The residual of the theoretical correction model, quantified by the CV RMS of the Ridge surfaces: 0.0033±0.0001 mag (NextGen $\log_{10} g = 1.5$), 0.0009±0.0000 mag (NextGen $\log_{10} g = 5.5$), 0.0010 ± 0.0000 mag (BT-Settl $\log_{10} g = 4.0$) and 0.0058 ± 0.0002 mag (BT-Settl $\log_{10} g = 5.5$), all, but one, evaluated for BP − RP ≤ 10. This term applies only to stars redder than the library sample, where the model correction is required. (iv) The uncertainty on the interstellar extinction. At fixed BP − RP the correction decreases as $\Delta = -(0.03282\, A_{K_s}^2 + 0.06366\, A_{K_s})$, so $\sigma_\Delta \simeq (0.06366 + 0.06564\, A_{K_s})\, \sigma_{A_{K_s}}$; for example, $\sigma_{A_{K_s}} = 0.05$ mag at $A_{K_s} = 0.2$ mag gives about 0.004 mag. For blue, low-extinction stars the budget is dominated by terms (i) and (ii),

TABLE 6 Coefficients of the 5th-degree ΔTESS polynomial (BT-Settl $\log_{10}$(g)=5.5), fitted for BP−RP ≤ 10 mag. Features were standardized and $\alpha = 1.0e-04$ was chosen by repeated shuffled 5-fold cross-validation; coefficients are given in the original (unscaled) variables.

Term	Coefficient
Constant (c_0)	-1.329229e-02
BPRP	1.593053e-02
A_K	-8.733715e-02
BPRP^2	-8.831643e-03
BPRP $\times A_K$	5.664925e-02
A_K^2	-2.357119e-01
BPRP^3	7.858785e-04
BPRP$^2 \times A_K$	-7.628333e-03
BPRP $\times A_K^2$	1.097561e-01
A_K^3	-3.082084e-01
BPRP^4	6.854960e-05
BPRP$^3 \times A_K$	-1.491960e-03
BPRP$^2 \times A_K^2$	-4.557967e-03
BPRP $\times A_K^3$	5.332775e-02
A_K^4	-8.391995e-02
BPRP^5	-9.747066e-06
BPRP$^4 \times A_K$	2.596816e-04
BPRP$^3 \times A_K^2$	-2.134561e-03
BPRP$^2 \times A_K^3$	1.324176e-02
BPRP $\times A_K^4$	-5.588693e-02
A_K^5	9.991291e-02

giving $\sqrt{0.005^2 + 0.002^2} \approx 0.0054$ mag. For red stars using the BT-Settl correction, terms (iii) and (iv) dominate, and the total uncertainty rises to about 0.008 mag at BP − RP = 7 − 10 mag. The error_correction function of GaiaXPy was unavailable for our custom passband, so the flux errors returned by GaiaXPy are likely underestimated; the magnitude error follows from $\sigma_mag = 1.0857\,\sigma_F/F$, and is added in quadrature to the terms above.

5 | APPLICATION TO TOI STARS AND DISCUSSIONS

In the broader literature, synthetic photometry from Gaia XP spectra routinely achieves a few percent accuracy (sometimes 1–3% or better after corrections), and even millimag accuracy after standardization for bands fully contained within the 330–1050 nm range (Gaia Collaboration et al., 2023; Munaretto et al., 2025). The accuracy derived here for the $\mathrm{TESS_{v2}}$ band, which slightly exceeds the red end of the Gaia XP coverage, is therefore competitive for stars bluer than BP−RP=3-5 mag and removes the colour bias. For reddened brown dwarfs an estimate of the interstellar extinction is required to achieve accuracy within 0.008 mag. The forthcoming Gaia DR4 release is expected to further improve the calibration.

We estimate that the improved $\mathrm{TESS_{v2}}$ absolute flux scale, in turn, increases the accuracy of several scientific outputs: planetary radii would change by about 1% on average ($\sigma \approx 2\%$), the flux-scale contribution to geometric-albedo uncertainties drops from 2–7% to below 0.5%, and the adopted quiescent flux, and hence the flare energy budget, shifts by about 6% for the very active star AU Mic while colour-dependent systematics are removed.

We recalculated TESS magnitudes for the stars in the TOI catalog[6] that have available XP spectra (see Fig. 10). We used GaiaXpy to obtain $\mathrm{TESS_{v2,cut}}$ magnitudes and, adopting the extinction values listed in the TOI catalog, applied the theoretical corrections derived from the BT-Settl $\log_{10}(g) = 4.0$ and $\log_{10}(g) = 5.5$ models. After excluding false alarms, false positives and ambiguous candidates (Guerrero et al., 2021), 5899 of the ≈7000 available data points remain. Their BP − RP colours have a mean of 1.031 mag (median 0.901 mag, $\sigma = 0.493$ mag); the resulting Δm values have a mean of −0.024 mag (median −0.030 mag, $\sigma = 0.057$ mag). A few sources show $\Delta m > 0.5$ mag (13 objects). All 13 stars lie within the validated Gaia XP magnitude range ($G \approx$ 12.5–15.3) and their RUWE is below the 1.5 quality threshold for all but two sources. No correlation was found between fraction of blended transits and Δm values. Only two of the 13 have a linked published work in SIMBAD.

5.1 | Implications for planetary radii

A change in flux density implies a change in the inferred bolometric flux and, if $T_{\rm eff}$ and distance are held fixed, in the stellar radius. Let's assume that the radius scales as the square root of the inband flux:

$$R_{*,\rm new} = R_{*,\rm old} \times 10^{-0.2\,\Delta m}, \qquad \Delta m = m_{\rm TESSv2} - m_{\rm TIC}. \quad (2)$$

Since the transit depth ($\delta_{\rm tr} \approx (R_p/R_*)^2$) is independent of the absolute flux zero-point, the planetary radius scales by the same factor, $R_{p,\rm new} = R_{*,\rm new}\sqrt{\delta_{\rm tr}}$.

By propagating the differences of magnitudes estimated for the TOI catalog into the planetary radii yields a mean change in radius sizes of 1.12% ($\sigma = 1.83\%$).

If stellar radii were derived from the TESS in-band flux, they would change by about 1%. Of course, this is not the case.

[6] TESS Objects of Interest Catalog for the TESS Prime Mission stars that have Gaia XP spectra.

TABLE 7 Transforming $\mathrm{TESS_{v2,cut}}$ magnitudes to $\mathrm{TESS_{v2}}$ magnitudes

Library	N_star	Transformation Eq.	σ
CALSPEC+SPSSV3.4	186	$\mathrm{TESS_{v2}} = \mathrm{TESS_{v2,cut}} - 0.0051\,(\mathrm{BP}-\mathrm{RP}) + 0.0031$	0.0116

1) The combined SPSS and CALSPEC libraries cover the range BP − RP < 5 mag, but there is only one point from 3 to 4 mag.

2) The constant term is included to null the median offset between the $TESSv2$ magnitudes from high-resolution and the $TESSv2_cut$ magnitudes from XP spectra. That is a standardisation term specific to Gaia DR3 BPRP spectra.

3) σ compares the high-resolution $\mathrm{TESS_{v2}}$ with the $\mathrm{TESS_{v2}}$ we reconstructed from XP spectra.

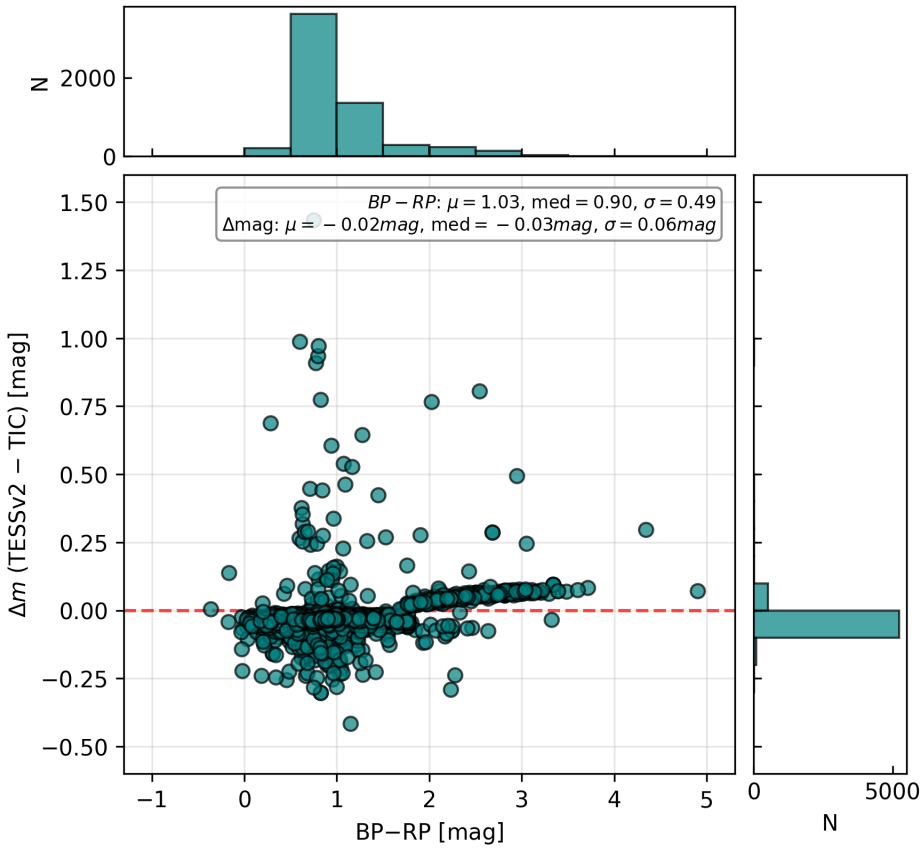


FIGURE 10 Δ TESS magnitudes versus BP − RP colors for ≈ 7000 TOI stars.

5.2 | Implications for geometric albedos

Accurate absolute fluxes are essential for converting secondary-eclipse depths into geometric albedos. The observed occultation depth can be written as:

$$\delta_{\mathrm{occ}} \approx \left(\frac{R_p}{R_*}\right)^2 \left[A_g \left(\frac{R_*}{a}\right)^2 + \frac{I_{p,\mathrm{th}}}{I_*}\right], \tag{3}$$

where the first term inside the brackets is the reflected-light contribution and the second is the thermal emission of the planet. While δ_{occ} itself is a relative quantity, the thermal term depends on the absolute stellar intensity I_*. Consequently, any fractional error $\varepsilon = \Delta I_*/I_*$ in the stellar flux appears directly in the recovered albedo:

$$\Delta A_g \approx -\left(\frac{a}{R_*}\right)^2 \frac{I_{p,\mathrm{th}}}{I_*}\,\varepsilon. \tag{4}$$

For a hot Jupiter such as WASP-18b (Arora & Goyal, 2024; Blažek et al., 2022), with $(a/R_*)^2 \approx 12.25$, a planetary temperature of 2500 K, and a stellar temperature of 6431 K, by assuming blackbody emission and integrating over the TESS

bandpass (Bessell & Murphy, 2012), we obtain a thermal contrast $I_{p,\mathrm{th}}/I_* = 0.011$. With a fractional photometric error of 0.15 in the stellar flux density, the resulting uncertainty in the geometric albedo is

$$\Delta A_g \approx 12.25 \times 0.011 \times 0.176 = 0.024,$$

corresponding to 7% of WASP-18b's albedo value. For Kepler-7, using the parameters from Demory et al. (2011), the same 0.15 fractional error in TESS flux density translates into a 2% variation in the derived albedo.

The $\mathrm{TESS_{v2}}$ magnitudes derived here reduce the typical absolute uncertainty from the ≈ 0.15 mag characteristic of the TIC broad-band estimates to a few millimagnitudes. The flux-scale contribution to the geometric-albedo uncertainty drops from 2–7% to below 0.5% and thereby strengthens the scientific return of secondary-eclipse and phase-curve observations performed with TESS.

5.3 | Implications for energetics of flares

Accurate absolute TESS magnitudes are also important for the characterization of stellar flares. The relative amplitude of a flare measured from a TESS light curve is independent of the absolute photometric zero point, but the conversion of this relative variation into a physical flare luminosity or emitted energy requires an accurate estimate of the quiescent stellar flux. An error in the absolute TESS magnitude propagates directly into the inferred flare energy. The TESSv2 calibration can be particularly useful for homogeneous studies of flare energetics across stars with different colours, where systematic errors in the adopted TESS magnitude could otherwise introduce colour-dependent biases.

AU Mic is an interesting example. TESS observations show frequent flares, with a mean relative flux amplitude of about 1% (Plavchan et al., 2020). For such events, an accurate quiescent flux is essential when converting the measured relative flare amplitude into an absolute flare flux and energy. The flare energy in the TESS band is obtained by multiplying the quiescent stellar luminosity by the equivalent duration of the flare, i.e. the time-integrated relative flare flux. For AU Mic the difference between the TIC and TESSv2 magnitudes corresponds to a 6.4% change in the adopted quiescent flux and therefore in the inferred TESS-band flare energies. While this shift is smaller than the 60% uncertainties that already dominate absolute energy estimates, a uniform absolute scale removes colour-dependent systematics that would otherwise appear when comparing flare energetics across stars of different spectral types.

6 | SUMMARY

We present a method to derive $\mathrm{TESS_{v2}}$ magnitudes using the Gaia DR3 XP spectra, an unprecedented dataset for this purpose. By employing the SPSSV3.4 and CALSPEC libraries of high-quality spectrophotometric standards together with a consistent zero-point calculation, we establish a traceable absolute scale for the $\mathrm{TESS_{v2}}$ magnitudes.

For CALSPEC stars with $6 < \mathrm{TESS_{v2}} < 16$ mag, XP-based $\mathrm{TESS_{v2,cut}}$ magnitudes agree with the high-resolution ones with a median offset of 0.0027 mag and $\sigma = 0.0050$ mag; the SPSS scatter is larger ($\sigma = 0.0134$ mag).

The missing red tail is corrected empirically for the library stars (adding $\approx$ 0.002 mag). These stars unfortunately span only a limited colour range. To cover red and highly reddened stars we therefore use ridge-polynomial surfaces in $BP - RP$ and A_{K_s} fitted to BT-Settl and NextGen models. In these surfaces the correction decreases quadratically with A_{K_s} (CV RMS = 0.0033, 0.0009, 0.0010, and 0.0058 mag for the NextGen $\log g = 1.5$, NextGen $\log g = 5.5$, and BT-Settl $\log g = 4.0$ and BT-Settl $\log g = 5.5$ grids, respectively). For stars bluer than $BP-RP \approx 3$ the empirical and theoretical corrections agree within the uncertainties. The total uncertainty is ≈ 5 mmag for blue, low-extinction stars and rises to ≈ 8 mmag for very red or reddened stars with $BP - RP \gtrsim 7.5$ mag.

For 5899 TOIs the new magnitudes differ from the TIC by -0.024 mag on average ($\sigma = 0.057$ mag). The ≈ 0.15 mag TIC uncertainty corresponds to up to $\approx 7\%$ in the geometric albedo of object like WASP-18b, and the TIC–$\mathrm{TESS_{v2}}$ difference to $\approx 6\%$ in the quiescent flux, hence the flare energies, of stars similar to AU Mic.

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This work has made use of the Gaia Spectrophotometric Standard Stars (SPSS) library (Pancino et al., 2012, 2021) and the CALSPEC spectrophotometric standard star library

maintained by the Space Telescope Science Institute (Bohlin, 2014, 2020; Bohlin, Krick, Gordon, & Hubeny, 2022). We thank E. Pancino for providing an updated version of the SPSS library (V3.4). We thank the Python, Astropy, NumPy, Pandas, PiVo, Sklearn, and Matplolib communities for providing their packages. This work has made use of the Python package `GaiaXPy` (Gaia Collaboration, Montegriffo et al. 2023). MM, and MP are members of the Gaia CU5 team. MM and MP thank Dr. Michele Bellazzini for his inspiring and constructive comments on synthetic photometry. MM and MG acknowledge the contribution of the ASI-INAF agreement 2021-5-HH.0. ("Partecipazione italiana alla fase B2/C della missione Ariel"). ARIEL (Atmospheric Remote-sensing Infrared Exoplanet Large-survey) is ESA's M4 mission, planned for launch around 2029. It will characterize the atmospheres of hundreds of exoplanets using transit spectroscopy in the infrared. MM thanks Dr. Schanche Nicole E. for information about the TESS passband release. MM and MG thanks Dr. Mattia D'Arpa for discussion on hot Jupiters.

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